

GuardGLP: Generative Meta-Models are Adversarial Classifiers



ddidacus/guard-glp

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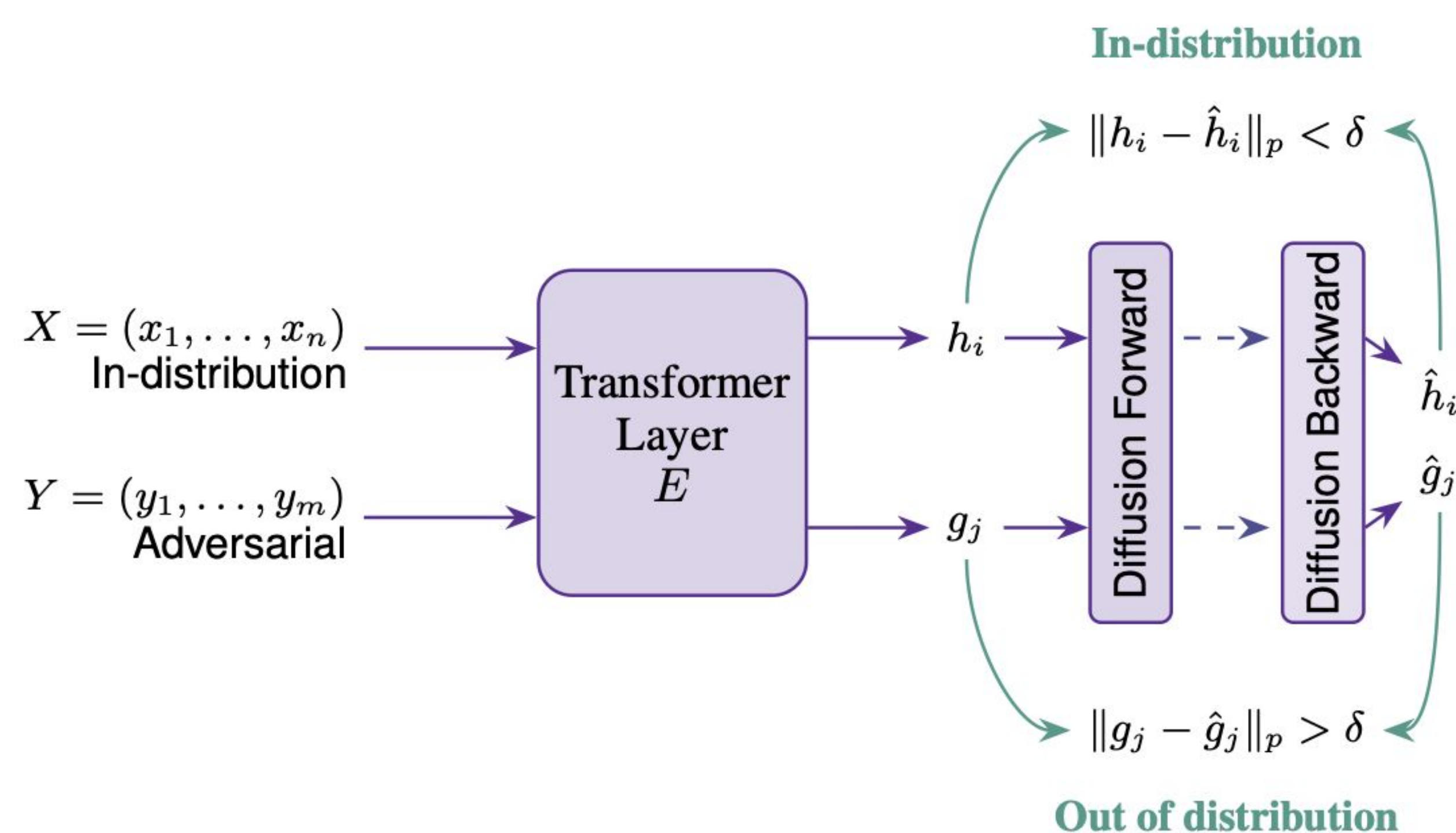
SUMMARY

- **Problem** : in language, adversarial queries can be hard to detect.

- ✗ No clear distance measure between inputs for semantics, characters
- ✗ High variety of attacks: data leakage, dangerous instructions, hate speech
- ✗ Generative models “shadowed” by classification methods, which poorly generalize across all categories of attacks

- **Proposed solution** : a whitelist model, adversarial attacks are off manifold

- ✓ **Generative Latent Prior** ^[1] : fitting the pre-train distribution in hidden space
- ✓ On-manifold = safe queries ; Off-manifold = unusual/unsafe queries
- ✓ We leverage diffusion anomaly detection: reconstruction/density based



Multiple formulations of Guard-Generative Latent Priors

Consider the diffusion latent: $z_j \sim \mathcal{N}(\sqrt{\bar{\alpha}_T} g_j, (1 - \bar{\alpha}_T)I)$
and reconstruction: $\hat{g}_j \sim \mathcal{N}(\mu_\theta(z_j, T), \Sigma_\theta(z_j, T))$

- Guard-GLP-RE: diffusion reconstruction error thresholding (p=2)

$$f^{\text{adv}}(g_j) = \mathbb{1}\{\|g_j - \hat{g}_j\|_p > \gamma\}$$

- Guard-GLP-DTE: noise posterior, kNN diffusion time estimation ^[2]

$$f^{\text{adv}}(g_j) = p(\sigma_t^2 | g_j) \propto \sigma_t^{-d} \exp\left(-\frac{1}{\sigma_t^2} \cdot \frac{1}{K} \sum_{g_i \in \text{kNN}(g_j)} \frac{\|g_i - g_j\|^2}{2}\right)$$

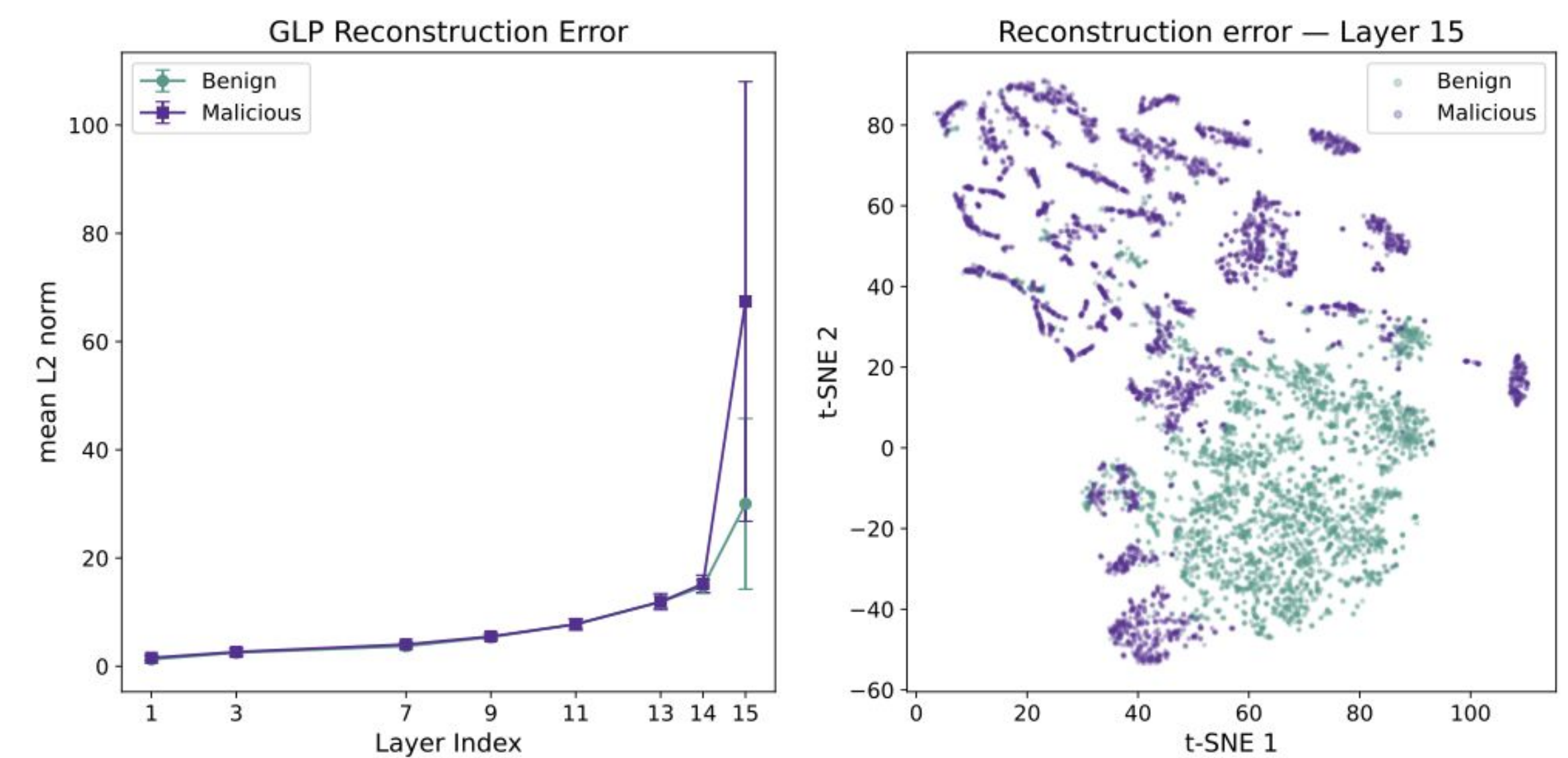
- Guard-GLP-TE: density estimation as in normalising flows

$$f^{\text{adv}}(g_j) = 1 - p_{\mathcal{D}}(g_j) \approx 1 - (p_z(z_j) | \det J_{T^{-1}}(g_j)|)$$

Baselines

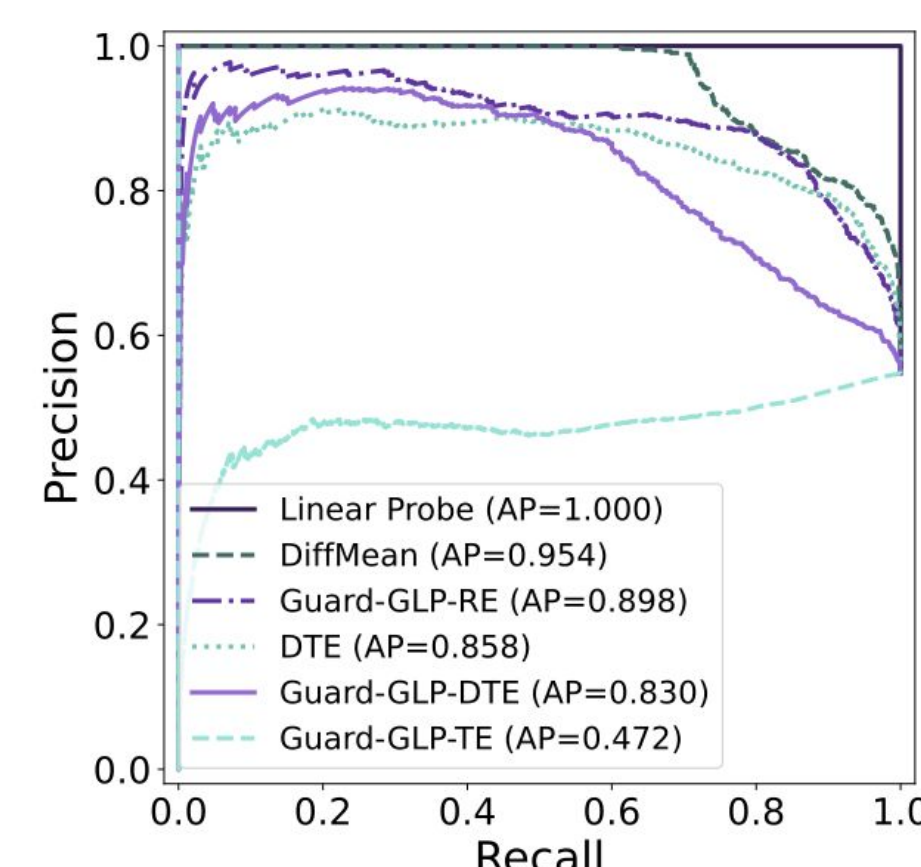
- Linear probing: $f^{\text{adv}}(g_j) = p(y_j^{\text{adv}} | g_j) = \sigma(W^T g_j + b)$
- DiffMean: $f^{\text{adv}}(g_j) = 1 - p(y_j^{\text{safe}} | g_j) \approx 1 - \left(\frac{g_j \cdot \bar{g}^{\text{safe}}}{\|g_j\|}\right)$

EXPERIMENTS & DETAILS

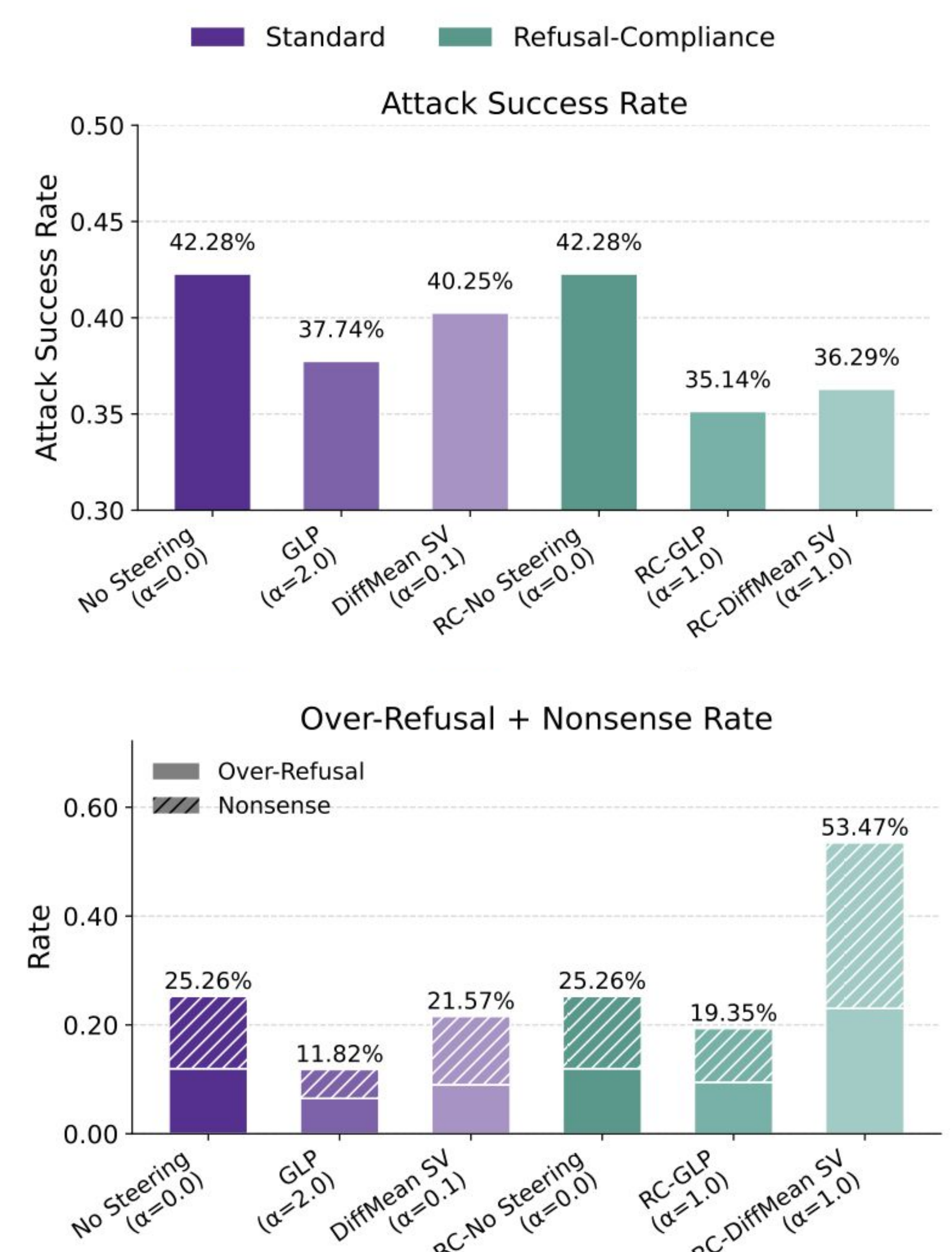


Token-level (GCG) attacks are off-manifold activations in GLP space. Reconstruction error explodes on the last layer.

Method	AUPRC (↑)	FPR (↓)	FNR (↓)
Guard-GLP-RE	0.89	0.19	0.15
DTE	0.86	0.24	0.16
Guard-GLP-DTE	0.83	0.10	0.43
Guard-GLP-TE	0.47	0.00	1.00
Linear Probe	1.00	0.00	0.00
DiffMean	0.95	0.01	0.30



GLP reconstruction error thresholding → best proposed classifier. Linear probe memorizes token-level features.



On manifold steering with GLP allows more stable and effective steering vector injection.

CHALLENGES

- Weak whitelist model → WildChat training
- Limited test set → extend to other attacks
- High variance for density estimation

[1] Learning a Generative Meta-Model of LLM Activations. Luo et al. 2026

[2] On Diffusion Modeling for Anomaly Detection. Livernoche et al. 2023